2019 8th International Conference on Affective Computing and Intelligent Interaction (ACII)

Facial Expression Recognition with Identity and Spatial-temporal Integrated Learning

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***Abstract*—Spatial-temporal structure of expression frames plays a critical role in the task of video based facial expression recognition (FER). In this paper, we propose a 3D CNN based framework to learn the spatial-temporal structure from expression frames for video-based FER. First, we use the data labeled with identities to train an identity network to capture the facial biometric features from expression frames. Second, we remove the impact of facial biometric features from the expression features and construct typical facial expression (TFE) features. Then, we feed the TFE features to a 3D network to discover spatial-temporal structure of expression frames. In the end, we feed the spatial-temporal vector to a fully-connected layer to get a vector for classification. The proposed method achieves comparable accuracy with the state-of-art of 88.54% on Oulu-CASIA, and is efficient to be used for the task of video-based FER.**

***Keywords*—Facial expression recognition, identity, spatial-temporal structure, 3D convolution**

I. INTRODUCTION

Facial expression recognition (FER) is the task defined to automatically classify the facial expression into several specific typical categories, e.g., sad, happy, surprise, angry, fear, disgust, etc. As FER is expected to estimate the emotional states and intentions conveyed by the facial expression of the subjects, it plays an important role in many applications of human computer interaction systems [1, 2].

The tasks of FER can be sorted into two groups according to the feature representations, i.e. FER for still image and FER for video or image-sequence. The FER technique for still images extracts expression features from the spatial structure of a single image, while FER for image-sequence recognizes the expression of subjects based on the features from image-sequence or video. Compared to FER on still image, FER for video or image-sequence recognizes the expression based on both the spatial structure and the temporal relation among continuous frames, and is feasible to capture the emotional states and intention of subjects during the entire video.

Some researches on FER on image-sequences use post-hoc integration methods to combine the features [3, 4], and obtain a suboptimal scheme to explore the spatial and temporal structure of expression characteristics. However, exploring integrated spatial-temporal features of expression for recognition in an optimized way is still an open question.

In this paper, we propose a 3D CNN based Network for image-sequence based FER. We employ a 3D CNN based module to explore the expression variation in spatial-temporal dimensions of expression video. Considering a typical facial expression expressed by distinct subjects (people) shows non-neglected difference due to their facial biometric shapes, the unique facial features of distinct person make the classification problem more complex, specifically in extracting spatial-temporal relationships. Removing the impact from facial biometric features may decrease the difficulty of facial expression recognition. Therefore, we train an identity network with the label of subject identities to extract facial biometric features from the face expression frames. We reduce the influence of facial biometric features by removing the extracted facial biometric features from the face expression features. In the proposed framework, we use a de-coupler to remove the impact of facial biometric features from the face expression features, and provide de-coupled expression features to a 3D CNN module to extract the spatial-temporal structure of expression.

Experiments results show that removing the facial biometric features from expression features can help the 3D CNN module to extract spatial-temporal structure and improve the facial expression recognition accuracy. The proposed method achieved comparable accuracy with the state-of-art of 88.54% in experiments on the dataset of Oulu-CASIA.

II. RELATED WORK

Tasks of FER can be classified into two categories: recognizing expression from still image and from image-sequence. Earlier research for FER mainly focuses on expression analysis based on still frames. As still image only provides spatial structure and constrains on facial

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expression at a static moment, the application of FER from still image is much limited.

In recent years, recognizing facial expression from video or frame sequences, as a dynamic and more natural event, is proved to be more effective to estimate the emotion state or intention of people [5-7]. Besides spatial information from images, temporal structure of expression is also appeared in image sequence. Extracting temporal feature together with spatial structure of facial expression in an integrated way is critical for the success of sequence-based FER task.

Many works based on traditional hand-crafted features have been conducted for all kinds of recognition tasks before deep learning is widely employed. In order to discover spatial information and temporal context in the meantime, these works extended traditional hand-crafted features to adapt to consecutive frames, such as 3D-HOG [8], LBP-TOP [9]. However, such methods only used shallow learning of FER, the feature's capability to represent complex variations over facial action parts is very limited.

Over the past few years, some research works applied deep neural network to extract temporal relationships and achieved promising results. The work of [3] utilized a deep network based on two different models to capture temporal appearance and geometry features respectively. The work of [4] proposed a deep evolutional spatial-temporal network (composed of PHRNN and MSCNN) to extract the partial-whole, geometry-appearance and dynamic-still information, and obtained a better performance than [3]. These works explored the temporal structure and spatial structure of facial expression frames in a parallel way, however, they provide inspirations to integrate spatial and temporal information with a unified optimization framework.

In recent years, 3D CNN has already shown its superiority as one of the state-of-the-art models in action recognition field as it processes both spatial information and temporal information in an integrated way. A relevant study in FER task using 3D CNN is 3DCNN-DAP [10] which incorporates a deformable parts learning component into a 3DCNN framework to capture spatial-temporal features from facial expression frame sequence. However, the huge amounts of parameters make the 3D convolution network easy to fall into overfitting, especially on a small database, which limits its applications.

In all of these works, inter-subject variability still makes facial expression recognition task a challenging question both in frame-based and sequence-based methods. Li et al. [11] proposed to use auxiliary identity information hidden in facial expression images to improve recognition accuracy in image-based FER in order to tackle the problem of inter-subject variability. They concatenated high-level emotion and identity features together as Tandem Facial Expression features for joint facial expression learning and outperformed the state-of-the-art methods. The success of [11] shows the potential of applying identity information to inter-subject variability problem in sequence-based FER tasks.

III. Our Approach

In our approach, we employ a 2D-3D CNN based network for video-based facial expression recognition in a spatial-temporal unified optimized way. We pre-train an Identity-network with labels and design a de-coupler to remove the influence of facial biometric features from expression features. We use a de-coupler to remove the influence of facial biometric features from expression features, and provide typical facial expression features for the 3D CNN network to extract special-temporal structure in a unified and optimized way. To decrease the amounts of parameters in the 3D CNN module (Temporal-network), we use features rather than raw video-frames as the input of module.

A. *Overview of the overall architecture*

As shown in Fig. 1, the proposed framework is mainly composed of three modules: Identity-network, Expression-network (shown as EXP-network in Fig. 1), and Temporal-network. Identity-network is trained with the label of identities and employed to extract facial biometric features from input facial expression images. Expression-network is trained to extract expression features. The proposed model uses a de-coupler to remove the influence of facial biometric features from the expression features and constructs the typical facial expression (TFE) features. Temporal-network is a 3D CNN based module to extract the spatial and temporal structure of TFE features.

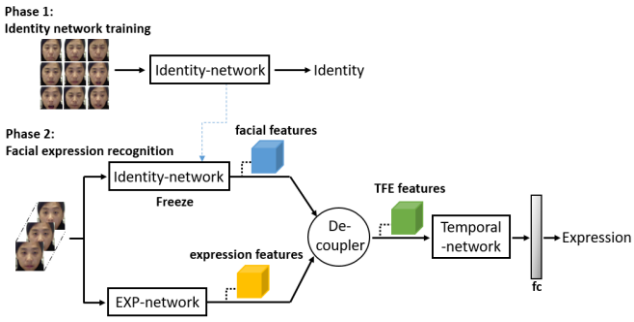

Fig. 1. Overview of the proposed framework. The blue, yellow and green volume represent output features from Identity-network, EXP-network, and Temporal-network, respectively.

The training process of the proposed architecture includes two phases. In Phase 1, we collect all images of each subject and pre-train the Identity-network with the label of their identities. The trained Identity-network is able to generate efficient facial biometric features for the recognition of subjects. In Phase 2, we freeze the parameters of Identity-network and merge the trained Identity-network with the other modules in the proposed network. Then, we train the structure with expression image sequences. The first two blocks of Expression-network module (EXP-network in Fig. 1) are used to provide description of expression features. The de-coupler removes the facial biometric features from the expression description, and constructs TFE features. After that, we feed the Temporal-network with TFE features of frames to learn the spatial-temporal structure among these frames, and form TFE vector. Then we feed the TFE vector into a fully-connected layer to get a vector corresponding to the label of class.

B. Identity-network

We use a ResNet-18 [12] based architecture to extract the biometric features of the subjects, and name it as Identity-network. The training process of Identity-network is named as Phase 1 where the whole Identity-network is trained from scratch with all subjects' images labeled with their identities. We use four blocks of ResNet-18 for Identity-network to recognize identities in Phase 1. The identity dataset is made by collecting all images in expression image sequences for each subject. We randomly select 80% images of each subject for training set, and the left for test set. After that, the first two blocks of the trained Identity-network are used to extract efficient facial features to differentiate subjects, and provide subject-specific description which influences the facial expression. In Phase 2, the output of the first two blocks of Identity-network is 128 feature maps in the size of 28 × 28 for each frame. These subject-specific descriptions will be subtracted from the expression features to remove the influence of biometric features from expression features.

C. Expression-network

Expression-network (EXP-network in Fig.2) is composed of the first two blocks of the ResNet-18 (until conv3_x layer). Expression-network is pre-trained on ImageNet and used in the proposed framework to extract expression feature from expression frame.

As a 2D network for analyzing single frame, EXP-network is employed to extract expression features from each single frame. The output of EXP-network consists of 128 feature maps with size of 28×28. The output is regarded as the expression features which are coupled with the facial biometric feature of subject. In order to remove the impact of facial features in expression description, we use a de-coupler to proceed a point-by-point subtraction operation on the corresponding channels between facial features and expression features.

We denote E (for expression description) and F (the facial biometric features) the two input of the de-coupler. The output, which is formulated as TFE, is the result of point-by-point subtraction of corresponding channels. The output value of unit at position (x, y) in the i-th feature channel, j-th frame can be formulated as Eq. (1).

$$TFE_{ij}(x, y) = E_{ij}(x, y) - F_{ij}(x, y) \tag{1}$$

After the de-coupler, we construct TFE features with 128 feature maps of 28 × 28 for each frame. The TFE features will be fed into a 3D convolutional network to extract spatial and temporal information in an integrated way.

D. Temporal-network

We use a 3D convolutional network-based module to extract temporal structure from TFE features, and name it as Temporal-network. Similar with 2D convolution, 3D convolution can be formulated as Eq. (2)

$$V_{ij}^{xyz} = \sigma\left(\sum_{m}\sum_{p=0}^{P_i-1}\sum_{q=0}^{Q_i-1}\sum_{r=0}^{R_i-1} w_{ijm}^{pqr} \cdot V_{(i-1)m}^{(x+p)(y+q)(z+r)} + b_{ij}\right), \tag{2}$$

where V_{ij}^{xyz} is the value at position (x, y, z) on the j-th feature map in the i-th layer. P_i, Q_i, R_i describe the sizes of 3D filters, and represent height, width , and temporal dimension, respectively; w_{ijm}^{pqr} is the (p, q, r)-th value of the filter connected to the m-th feature map in the $(i-1)$-th layer. The activation function $\sigma(x) = \max(0, x)$ is a nonlinear operation applied in our model, which is also called ReLU [13].

We adopt conv4_x and conv5_x of 3D-ResNet18 [14] pre-trained on Kinetics for the Temporal-network. As shown in Fig.1, the 3D convolution operation in the proposed framework is placed next to two deep 2D CNNs (Identity-network, and EXP-network). There are two reasons for this arrangement. First, a deep 2D CNN is able to extract high-level spatial features, which helps 3D CNN to extract spatial-temporal features efficiently. Furthermore, it also helps decrease the amounts of parameters required in 3D CNN, and is helpful to avoid overfitting in small database. We design a 3D average pooling layer at the end of Temporal-network with the kernel size of 2×7×7 and obtain a 512-length output vector. Finally, we feed the output vector to a fully-connected layer to get the classification vector. The fully-connected layer is applied with the size of 512 × 6 for a six-classification task.

IV. EXPERIMENT

We evaluated the performance of the proposed method in terms of classification accuracy on a well-known benchmark database: Oulu-CASIA database [15]. We compared the performance of our approach with other state-of-the-art algorithms for FER based on image-sequence. Moreover, we evaluated the performance of our model on Oulu-CASIA by confusion matrix.

A. Dataset

The Oulu-CASIA dataset includes 2,880 image sequences collected from 80 subjects labeled with six emotion labels: anger, disgust, fear, happiness, sadness, and surprise. Each of the videos is captured with one of two imaging systems, i.e., near-infrared (NIR) or visible light (VIS), under three different illumination conditions. Typically, 480 videos collected by the VIS System under normal indoor illumination are employed for 10-fold cross validation experiments. We divided the 80 subjects into ten groups by their IDs in ascending order to avoid subjects appearing in both the training and testing sets simultaneously.

B. Preprocessing and data augmentation

We detected faces in the input image sequences using SDM algorithm [16] which can extract 49 facial landmarks, including two eyebrows, two eyes, a nose and a mouth. These landmarks were detected in order. The position and region of local parts were also detected via this method. The detected faces were then cropped, and rescaled to a fixed size of 244×244. We adopted online augmentation strategy with two data augmentation techniques in both two datasets: random cropping and random horizontal flipping. Firstly, we extracted a region of 224×224 randomly from the rescaled images when training, and a 224×224 region was cropped in the center of images when testing. After that, the whole image sequence was horizontal flipped with a probability of 0.5. The horizontal flipped augmentations were only applied in training phase.

C. Experiment Setting

In the proposed architecture, we selected 7 frames rather than all frames from each image sequence to feed into the network in order to improve computing efficiency. We divided the image sequence to 7 segments of equal durations. Then we randomly chose one frame from each segment during training. The middle frame of each segment was selected when testing. For model training, we used the mini-batch stochastic gradient descent algorithm to learn the network parameters, where the batch size was set to 16 and the momentum was set to 0.9. The learning rate was initialized as 0.001 and decreased to its 1/10 every 70 epochs. The whole training procedure stopped at 200 epochs. Considering the Oulu-CASIA dataset contains hundreds of image sequences, to avoid overfitting, we set the weight decay to 0.1.

D. Experiment result

We compared the performance of our proposed method with the state-of-the-art methods on the Oulu-CASIA dataset in Table 1.

Table 1 shows our method obtained performance with accuracy of 88.54%, which is comparable with the state of art, Kuo et al. [17]. Except for the work of [17], our method achieved around 2% accuracy higher than other methods. As the Oulu-CASIA dataset doesn't provide parameters for train and test split, direct comparison on test accuracy may not very convincible in the evaluations of different algorithms. The better way to evaluate the performance of our method on Oulu-CASIA is to proceed cross validation. Table 2 show the confusion matrix for Oulu-CASIA. In Table 2, An, Di, Fe, Ha, Su and Sa represent angry, disgust, fear, happy, surprise, and sad, respectively. The labels in the leftmost column and on the top represent the ground truth and prediction results, respectively. The number of test videos is 480 all over the 10-fold cross validation, including 48 videos for each label.

TABLE I. OVERALL ACCURACY ON THE OULU-CASIA DATASET

Comparison on Oulu-CASIA		
Method	***Descriptor***	***Accuracy***
Guo et al. [19]	Atlases	75.52%
Yu et al. [18]	DCPN	86.23%
Jung et al. [3]	CNN-DNN	81.46%
Zhang et al. [4]	PHRNN-MSCNN	86.25%
Kuo et al. [17]	CNN	91.67%
Our method	2D-3D CNN	88.54%

TABLE II. CONFUSION MATRIX ON THE OULU-CASIA

	An	Di	Fe	Ha	Su	Sa
An	**82.5**	8.75	2.5	3.75	0	2.5
Di	13.75	**81.25**	1.25	0	0	3.75
Fe	0	2.5	**88.75**	3.75	2.5	2.5
Ha	0	1.25	1.25	**97.7**	0	0
Su	1.25	0	7.5	0	**91.25**	0
Sa	6.25	2.5	1.25	0	0	**90**

Table 2 shows our algorithm performed well in recognizing anger, happiness, fear, sad and surprise. We observed that the accuracy for disgust was much lower than other emotions. Most failure cases labeled with disgust were recognized as angry. The results indicate that a variety of facial expressions in Oulu-CASIA labeled as disgust were similar to anger.

In summary, experiment results show our approach is an effective and generic solution for video based facial expression recognition.

V. CONCLUSION

In this paper, we propose a 3D CNN based architecture for the recognition of expression in video. We explore the identity information (facial biometric feature) hidden in facial expression to remove the influence from facial features in expression features. Moreover, we build a deep network including 2D and 3D convolution to integrate spatial and temporal information in a unified optimization structure. Experimental results show the efficiency of the proposed method. Our approach is an effective and generic solution for video based facial expression recognition.

ACKNOWLEDGMENT

This research was funded by the Science of Technology Program of Guangzhou City (201903010040), China.

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